AI-Powered Retail Shelf Monitoring and Stock Availability Detection

P. Kumar
Dept. of Computer Science and Engi neering
Rajalakshmi Engineering College Chennai,India
kumar@rajalakshmi.edu.in

S. Aravinthaa
Dept. of Computer Science and Engi neering
Rajalakshmi Engineering College Chennai,India
230701032@rajalakshmi.edu.in

D. Karneesh
Dept. of Computer Science and Engi neering
Rajalakshmi Engineering College Chennai,India
230701141@rajalakshmi.edu.in

Abstract While customer satisfaction is strongly related to the product being available on the shelves, the efficiency of the shelf replenishment and the accuracy of inventory is equally crucial, especially when assessing the shelves' condition via a traditional inventory record, which does not directly describe the physical condition and thus is not a direct measure of satisfaction. In addition to the above, computer vision-based monitoring systems for retail products present multiple challenges to the visual information such as occlusion, product overlapping, similar packaging, lighting changes and misplaced products. This paper proposes a framework of Artificial Intelligence (AI) powered Retail Shelf Monitoring and Stock Availability Detection system that uses the computer vision technology based on camera, product detection technology based on YOLO, shelf region identification technology, product counting technology, weight measurement technology based on IoT sensor, inventory/Point-of-Sale (POS) information and centralized monitoring. It saves the bounding box, product class and confidence score for each product detected, and records to a shelf region the product detection, and estimates how much product is visible. A complementary quantity is determined by weighing with the load cells in the shelf, and, by knowing the weight of a single product. Visual and physical estimates are compared, and potential errors that may occur (such as occlusion, counting errors, misplaced products, sensor abnormalities and inventory discrepancies) are identified. The information generated is associated with minimum-stock levels and the information about inventory is used to find if a product is in OUT-OF-STOCK, LOW-STOCK or NORMAL condition; If exceptions are detected, an automated alert is sent to the store staff. The proposed framework does not assume that visual, physical and inventory information are mutually exclusive, rather than object detection being the ultimate solution. At this point, the actual project and experimental data are not available, and hence the paper introduces an evaluation methodology, test cases, a mathematical evaluation metric and the result tables in which the experimental values are clearly marked.

Keywords: *Retail shelf monitoring, product counting, stock availability, multimodal sensing, product detection, computer vision, YOLO, IoT, inventory monitoring, out-of-stock detection.*

I. INTRODUCTION

To ensure customer satisfaction and inventory records are updated automatically. Having a stock record keeping system that is not in order and having an empty or emptying shelf are not necessarily the same problem and are not just a merchandising or display problem. Ideally, a retail store would have a way to know the exact amount and location of each product on each shelf at all times, notice when there were any discrepancies and have the correct information in front of staff before it reaches the customer. In practice, it has been found that the function of a bar code or a point of sale system is to describe the transactions of a product and not the actual placing of products on a particular shelf. This affects an actual inventory system. The other challenges with manual inspections include large shelves, multiple product types, product density and frequently changing displays. Computer vision is a natural approach for observing the status of the shelf. Stockport et al. have previously investigated various techniques for visual detection and recognition of grocery products such as local image features, Bag-of-Words representations and deep neural networks, and have found that problems in recognizing products in a supermarket scene, as opposed to isolated products, include the difficulty in identifying products that are visually similar, partially occluded, rotated, densely packed and under different illumination conditions [1]. Identifying "densely-packed products" (where many items of similar appearance are stacked side-by-side). This process is further explained by the introduction of large-scale retail benchmarks such as the Sugi-110k [2]. The fine-grained recognition and realistic retail scenarios [3, 4] could also be worked thanks to the product-recognition datasets such as RPC, Products-10K. More recently, deep-learning-based methods have yielded more robust methods of detecting and recognising products directly from images taken of the shelf. For example, the deep learning pipeline for intelligent retail environments proposed in the above-

mentioned study first identified the product on the shelf, then made an embedding-based recognition phase to establish a connection between the recognized product and the SKU level identity [5]. Research shows that deep learning and object detection are crucial for product-packaging recognition in retail images [6]. But product detection is not a full solution to the stock problem. A detector can detect visible products, but if there are other products covering up the products or if packages are lost, then a visible count may not equal the actual physical count, and if a product is in the store but not on the assigned shelf, it will not be detected. Empty-shelf-detection is therefore a related, but distinct problem. For empty/shelf analysis, some studies have been carried out based on the graph method, on the modeling of superpixels and their relationships [7] and some deep-learning approaches have been applied, to detect empty or frontal OOS on shelf images [8]. Other studies have focused on the problem of out-of-stock detection in egocentric shopping cart images with the aim of providing visual models for identifying the out-of-stock items to be restocked [9]. One way is to extend the vision-based autonomous sensing system to mobile supermarket inspection [10] to autonomous inspection of shelves based on vision. By research, the combination of product availability with product placement is possible, as well as products and racks detection by object detection and layout matching [11]. These approaches are suitable starting points of automated retail monitoring and are predominantly based on the visual domain. It's not a simple task to make a camera reliably capture all of the back-views of the product and an object detector determine if a product is out of stock or if it is temporarily out of sight but is actually in stock another location or is misrepresented in the stock system.Weight sensing, on the other hand, will give the physical proof, but typically requires additional data to identify the product and the position of the product on the shelf. While POS and inventory systems give transaction level information, it does not necessarily reflect the actual condition of a shelf. Where more than one sensing source is used to corroborate the other, and not to rely on any single one as a full picture of the availability of the shelf. The paper's proposed framework thus combines product detection by an image sensor with shelf-region assignment, visual product counting, IoT-based weight sensing, inventory/POS information, discrepancy analysis, and generation of alerts automatically. It provides mainly: (1) a unified workflow for product detection and stock status monitoring at the shelves; (2) mathematical integration of visual product quantity and stock status derived from the product weight; (3) a discrepancy mechanism to detect potential occlusions, counting mistakes, misplaced products and unexpected sensor abnormalities, and to detect stock status mismatches; (4) methodology that allows the complete system to be assessed without assuming experimental results that have not yet been measured. The arrangement of the paper is as follows. Section 2 summarizes related research from the retail product recognition, retail shelf monitoring, out of stock detection, and deep-learning-based retail vision perspectives. The proposed multimodal framework and the mathematical formulation of it are described in section 3.Section 4 explores a practical approach to implementing a model, encompassing hardware, software, model training, integration with IoT, and database services. Section 5 specifies the experimental evaluation method and the tables of expected results. The conclusion and directions for future development are provided in Section 6.

II. LITERATURE SURVEY

Research on automated retail monitoring has been undertaken in several directions, some of which are partially related, such as product recognition, dense object detection, shelf-gap detection, planogram analysis and out of stock identification. Early supermarket product-recognition research presented the problem of recognizing grocery products directly from shelf images as a challenge due to the significant variation of food packages and similarity of the visual features of different food products [1]. The Grozi-120 dataset and grocery-image benchmarks were useful for exploring recognition when products are not shown as single catalogue images, as is needed for realistic shelf conditions. Subsequently, as convolutional representations made it possible to learn visual patterns. The RPC dataset focused on another practical use of retail vision, namely a large-scale checkout oriented dataset consisting of single and multi-product images, suitable for studying product recognition in realistic retail environments [3]. Products-10K also highlighted the fine-grained recognition of product categories with similar-looking packaging, by offering a diverse set of product categories with similar-looking packaging [4]. While these datasets aid in identification, they differ from the one we are examining here; our focus is on the number of physical units on specific shelves, rather than product categories. A challenge is added due to the dense shelf detection. The authors of Goldman et al. were particularly looking for an object detection system to be able to cope with densely-packed retail environments, where many items are placed in close proximity, often with similar appearances [2]. The proposed system is highly relevant to the data set as

this is necessary for estimating the number of shelves in the monitoring area prior to the counting process. The heavy-duty SKU-110K however is mainly designed for retail object detection and cannot be used as the sole weight or state-of-the-pos detection for multi-modal inventory verification. Object-detection architectures have undergone significant development over time, paralleled by retail research. Methods based on the region also showed that proposals learned from the regions could be very useful for accurate object localization, as exemplified by Faster R-CNN [12]. YOLO deal with performance detection for real time applications [13]. The YOLOv4 architecture and training technique introduced several new changes to maintain the tradeoff between speed and accuracy in real-time object detection [13] and the YOLOv7 further enhanced this capability for real-time object detection [14]. The proposed framework is not version-specific for YOLO as the specific YOLO version can be chosen based on the final hardware, dataset, and accuracy requirements and the deployment constraints. Research on recognition in the retail sector is no longer limited to simple bounding-box detection. In [5] Pietrini et al. introduced a two-step method in which the products detected are encoded as embeddings and compared to each other using similarity measures and are recognized at the SKU-level. This is helpful if the number of product classes is vast and products visually vary at a fine level. Santra et al. studied part-based, annotation-free fine-grained recognition, and showed that discriminative product regions can be helpful when products are only slightly different in their visual aspects [15]. It has also been proposed to employ data augmentation since collecting and annotating all possible package orientations is costly; use of synthetic multi-angle product images has been explored to enhance grocery-product recognition in the presence of limited training data[16]. These studies have shown that a fine-grained appearance, viewpoint, occlusion, and limited labelled data are important features that must be addressed by a visual component of a retail system. Another direction is to study shelf localization and empty-space analysis. Santra et al. [7] introduced an image representation model based on graph to detect gaps in supermarket shelves. This will be helpful in situations where the desire is to locate areas with empty shelves even when single product detection is challenging. The direct detection of out-of-stocks by deep learning has also been examined. Allegra et al. applied egocentric visual data to identify regions which indicated out-of-stock situations to enable the employee to intervene [9]. De Simone et al. designed an autonomous mobile robot for supermarket shelf analysis, which is the combination of the robot navigation with the deep learning technique to detect empty and partially empty shelves [10]. Increased attention has been paid to better out-of-stock detection in recent studies, which have included fully empty shelf areas as well as frontline out-of-stock scenarios, reflecting the fact that the information for an occupied shelf can be insufficient to indicate that a product is in stock [8]. Another aspect of shelf monitoring that is linked to planogram is the fact that a product can be on the shelf but in the wrong shelf position. Saqlain et al. proposed a hybrid technique, which was a combination of product detection and comparing with a planned layout, which is called a planogram, to measure the shelf monitoring and planogram compliance[11]. This is a type of work that could be useful in the proposed framework for misplaced-product detection because product availability is in relation to product location, but not necessarily the same. The number of these products cannot be used to identify the item's position.POG systems do not necessarily show the actual presence of items located behind the visible ones on the shelf with precision. It is evident in the literature that there is a recurring dichotomy between visual recognition and shelf gap detection, compliance of layout and inventory information. Visual approaches can show direct evidence of what the camera is seeing, however with occlusion, insufficient lighting, high product density and similar package types, there may be a decrease in reliability of the visual approach. When there are multiple product types in a sensing region, weight-based sensing can only offer physical evidence of the mass of that particular product at that point on the shelf and is generally unable to determine which product in that region, by type, is contributing to the mass. POS systems can give transaction information but will not necessarily suggest what is happening on the shelf.

This research gap is not lack of retail object detection but it is related to the integration and cross-verification of multiple evidences at shelf level. Some existing research has shown that deep-learning detectors can be used to detect products and empty spaces of shelves, and other systems can be developed to solve the problem of planogram compliance, or for autonomous inspection of the shelves. The proposed framework takes this direction introducing a common decision making layer that enables a comparison between visible product quantity, weight-based quantity and inventory stock status. This design aims to minimise reliance of a single sensing modality and to create a more intelligible basis to detect potential discrepancies in the stock.

III. PROPOSED WORK

III.I System Overview

The system utilizes shelf images or videos captured by cameras. Measurement data are obtained from IoT-based weight sensors. It incorporates inventory information to help verify the detected stock levels. From each input image, the detected products are represented as:

pd=[bi,ci,si] i=1 . . .n (1)

where

bi boundary area
ci casual area
si segmented area

The confidence value is in [0,1] and only detections for which the confidence value is ≥ "threshold" (τdet) are accepted. The system then assigns products to the shelf region, and performs visual quantity vs weight quantity and inventory checks. It facilitates the recognition of the normal stocks, low stocks, out-of-stock, misplaced stock and possible discrepancies in inventory.

III.II Data Acquisition

Images or videos are captured from fixed cameras in the shelves. The physical weight of products on the shelf is measured by the IoT sensor, such as load cells. The sensor readings are sent to the processing system by an ESP32 microcontroller. The recorded stock can be further referenced by Inventory/POS information. Each observation is a record of the shelf identifier, product information, measurement by one of the sensors, and the time of the measurement.

III.III Image Preprocessing

Images captured are resized and normalized prior to detection. Also during training, image augmentation techniques are employed to account for variations in lighting conditions, product orientation and partial occlusion.

III.IV Product Detection

A YOLO-based object detection model is used to detect and classify retail products. For every detected product, the model returns the bounding box, product class and confidence score. The detections below the threshold τdet are not accepted for counting.

III.V Shelf Region Identification

The shelves are subdivided into pre-defined regions Rk. The location of the bounding-box of detected products is used to assign the products to the corresponding shelf region. This makes it possible to count products based on their shelf position and to identify products which are put in the wrong position on the shelves.

III.VI Product Counting

The amount of product p in shelf region k is calculated as the following:

V(k,p)=∑i1[ci=p AND si≥τdet AND bi∈Rk]. (2)

This indicates the number of products which are visible in the detected product area on a given shelf.

III.VII Weight-Based Stock Estimation

The weight of the physical shelf is given by the IoT sensor. The quantity of the product is estimated to be

W(k,p)=Mk−Mktarewp, (3)

In which, Mk is the measured weight, and Mktare is the weight when the shelf is empty.In which, Mk is the measured weight and Mktare is the empty-shelf weight and wp is the unit weight of the product.

In the proposed system, the number of products detected from the shelf image is compared with the quantity estimated from the weight sensor. The difference between these two values is calculated as

Δ(k,p)=|V(k,p)−W(k,p)|\Delta(k,p)=|V(k,p)-W(k,p)| (4)

where V(k,p)V(k,p) represents the visually detected quantity and W(k,p)W(k,p) represents the quantity estimated from the measured weight. A larger difference may occur when products are partially occluded, incorrectly counted, placed in another shelf position, or when the weight sensor does not provide a reliable measurement. The system therefore uses this difference as an additional indicator for identifying possible stock inconsistencies.

The stock information retrieved from the inventory also comes into consideration for finalizing the stock.

III.VIII Low Stock / Out of Stock Detection

The stock condition category is considered to be

S(k,p)={OUT-OF-STOCK,Q(k,p)=0LOW-STOCK,0<Q(k,p)<Q(k,p)minNORMAL,Q(k,p)≥Q(k,p)min. (5)

Where
Q(k,p) final decision quantity
Q(k,p)min indicates the low level.

III.IX Misplaced Product Detection

The detected product class is checked against the expected products of each shelf region. A product that is found in an unwanted area is marked as a misplaced product. A mapping from the shelf and the product is kept in the inventory database as expected.

III.X Alert Generation

The system sends alerts for out-of-stock, low-stock, misplaced products and big differences between visual and weight products. All alerts include: Product Name, Shelf Number, Quantity, Discrepancy Value, Timestamp, and Inventory Status. Information is centrally saved in the database and present in a monitoring dashboard.

This algorithm is designed to detect the availability of products in the retail shelf.

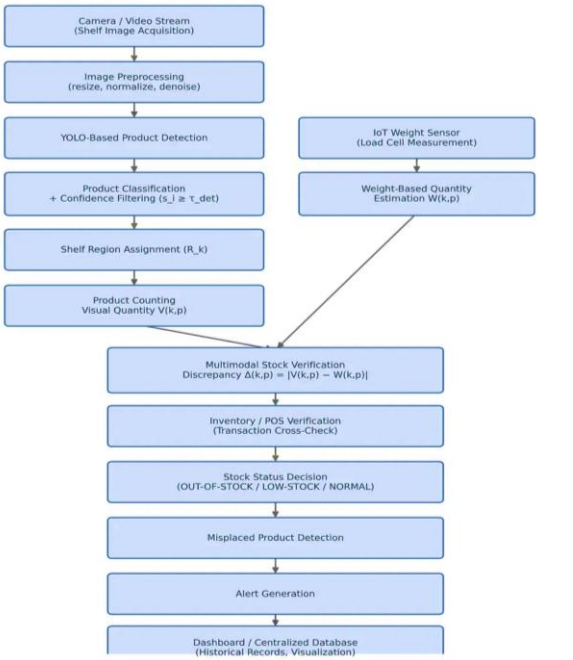


Fig. 1. System Architecture of the Proposed System

IV. IMPLEMENTATION

IV.I Dataset Preparation

The images in the dataset are taken from retail shelves and product categorized the conditions of the shelves. Class labels, Bounding boxes are used for each product. The prototype contains 3000 images of 10 different product categories and around 15000 product instances are annotated. The dataset samples are divided into 70% training , 15% validation and 15% test. The images include variations in lighting, product arrangement, partial occlusion, and ,product orientation . Other public retail datasets like SKU-110K, Products-10K, and RPC can also be used to support model development and comparison [2]–[4].

IV.II Model Training

The retail shelf dataset trained by the YOLOv8-based detector. The number of training epochs is 50, the batch size is 16 and the image size is 640 × 640 pixels, the batch size is 16. Data augmentation techniques such as brightness adjustment, horizontal flipping, scaling and cropping were employed in the training of the deep learning models. Assign the initial value of the threshold detection is set to 0.50.

IV.III IoT Sensor Integration

Weight layer continuously provide weight measurements at the IoT sensor. Where they are used along with the visual detection results for stock verification. This prototype is based on a 5 kg load cell, an HX711 amplifier and an ESP32 microcontroller. The tare value Mktare is taken when the shelf is empty, and the unit product weight wp is taken from the product reference information. Samle of Sensor data is approximately once every five seconds. A simple calibration technique designed to reduce variability due to product handling. Poor readings and errors in readings are not considered to represent a shortage of goods, but are input and recorded separately.

IV.IV Database and Dashboard

Shelf identifiers, product identifiers, timestamps, visual counts, physical quantity estimates, discrepancies, inventory data, stock classification, sensor status, generated alerts are all stored in the MySQL database. The dashboard gives a "flavored" view of the shelf level summary and facilitates operators identifying products that need replenishment. The database contains logical tables for Products, Shelves, Detections, SensorReadings, InventoryRecords, StockStatus and Alerts. When integrated with operational retail information, authentication and access control may be used.

V. RESULTS

The data in this section is not measured experimental data or implemented data that was reported so that it would be a methodology for evaluating instead of fabricated numerical results. For a multimodal stock-monitoring system, the proposed evaluation should be conducted at several levels as a single metric cannot capture the performance of the system. The visual detector should be assessed for standard detection metrics like precision, recall, F1-score, mAP at IoU=0.5 and mAP over an appropriate IoU range. In the case of predicted detections, Precision is TP/(TP+FP), Recall is TP/(TP+FN), and F1-score is 2(Precision×Recall)/(Precision+Recall). A predicted box is treated as a correct detection when its overlap with the corresponding ground-truth box is at least 0.50. The predicted bounding boxes are matched with the corresponding ground-truth boxes, and detections with an IoU value of 0.50 or higher are treated as correct detections. The resulting precision, recall, and detection accuracy are then calculated based on these matches. Secondly, performance for the product should be assessed separately from the detection performance because the detector might get good localization performance, but give the wrong number. Count error for shelf region k and product p can be written as Ecount=|Qtrue−Qpred| and when Qtrue>0, normalized count error can be computed. Third, known product values at the sensing site should be used to evaluate the weight-based estimator. MAE = (1/n) X j=1n | Qjtrue−Qjweight| can be used as an MAE, whereas percentage error can be computed only when there is a meaningful denominator. Fourth, the multimodal verification should be validated on

controlled cases where the number of items and the number of items counted visually are manipulated. These cases should cover normal shelves, actual blank shelves, low stock shelves, partially obstructed items, misplaced items, visual false detection, sensor anomalies and stock discrepancies. Fifth, recall, precision, false-alert rate and missed-event rate should be used to assess alerting performance at the level of individual events.

VI. Conclusion and Future Work

The proposed AI-Powered Retail Shelf Monitoring and Stock Availability Detection framework considers retail shelf monitoring as a multimodal inventory-observation problem, not just an object-detection problem. The framework uses product detection based on YOLO technology, shelf-region ID and visual counting for products, with an independent physical shelf quantity and inventory estimation provided by the IoT-based weight sensor, and a transactional shelf quantity and inventory estimation provided by the product information. This mathematical formulation maintains the separation between visual weight-based quantity W(k,p), quantity V(k,p) and their discrepancy Δ(k,p) . The stock-status layer then offers a simple classification of the state of the stock as OUT-OF-STOCK, LOW-STOCK, and NORMAL, and the misplaced-products and alert mechanism adds to the monitoring by reaching into the actionable realm of shelf management. While individual components like retail product recognition, dense shelf detection, empty-shelf analysis, out-of-stock detection and planogram monitoring have been shown to be feasible, there are further implementation and evaluation requirements associated with integrating visual, physical and inventory evidence. The current paper, therefore, does not make any claims of experimental superiority or report performance without measurement. The second phase of the project should involve gathering and annotating the Project specific data set, deploying the selected YOLO model, fine-tuning the weight-sensing layer, creating a camera-sensor synchronization data set, designing the final end user inventory/POs interface, and testing the system for the project specific test cases and metrics defined in Section 5. The system can be further improved by adding multi-camera tracking, better SKU recognition, and sensor error detection. Future work may also explore edge-based processing, adaptive discrepancy handling, automated replenishment, and planogram integration. The accuracy of detection, the error of counting, the classification of stock status, the latency, the number of data sets, and the performance of the sensor should be replaced by the actual experimental values only after the controlled experiments are completed..

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